

FLUOR

ANCHOR BOLT DESIGN CRITERIA

PURPOSE

This practice establishes guidelines for the design of headed type anchors into reinforced concrete foundations.

SCOPE

This practice includes the following major sections:

- NOTATION
- DUCTILE DESIGN PROCEDURE AS DESCRIBED BY ACI (AMERICAN CONCRETE INSTITUTE)
- GENERAL
- REFERENCES
- ATTACHMENTS

This practice covers the design procedures outlined by ACI (American Concrete Institute) 349, Appendix B. The procedures of the UBC (Uniform Building Code) 1925.3 are provided in Attachment 02. Each consists of 2 parts: design of steel headed anchors and design of the concrete embedment. Related dimensional requirements for the design are included in appropriate sections of the practice.

APPLICATION

The approach to be followed will be determined at the beginning of each project. Each method must be used in its entirety. Steel anchors and concrete embedments must be designed according to the same method.

This practice applies to headed anchor bolts and threaded rods with tack welded nuts. Where other anchor systems are utilized, this practice may serve as a guideline.

Ductile design of anchors is preferred for designs in UBC defined seismic Zones 3 and 4.

Ductile design as prescribed by ACI 349 will be followed when designing nuclear facilities.

Design limits less conservative than those specified herein may be used with prudent engineering judgment.

In addition to the requirements of the body of this practice, refer to Attachment 06 for special limitations when designing tall vertical vessels. Investigation of overlapping stress cones and intersections of edges should be considered with the design of vertical vessels.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

ACI 349, Appendix B, will be referred to as ACI throughout this practice, unless another ACI publication is specifically named.

UBC 1925.3 will be referred to as UBC unless another section of the code is specifically named.

Galvanized anchor bolts will be provided for all exterior and corrosive interior applications unless dictated otherwise by the client or the job site location. To ensure ductility, use the design values specified in the table for galvanized anchor bolts. Where plain anchor bolts are specified for exterior use or installation in a corrosive interior atmosphere, the reduced design values specified in the table for nongalvanized anchor bolts will be used.

NOTATION

Symbols

A_b	Tensile stress area (square inches) of a bolt or stud.
A_p	Projected area (square inches) of an assumed failure cone or truncated pyramid. The cone or pyramid radiates from the bearing edge toward the free surface at an angle of 45 degrees.
A_r	Reduction of the projected area (square inches).
A_{sb}	Area of reinforcement required by design for the lateral bursting failure mode (square inches).
A_{st}	Area of reinforcement required by design for the tension failure mode (square inches).
A_{sv}	Area of reinforcement required by design for shear failure mode (square inches).
a	Out-to-out dimensions of bearing edges. Refer to Attachment 09, Figure 1.
b	Out-to-out dimensions of bearing edges. Refer to Attachment 09, Figure 1.
B'	Ultimate bursting design load (kip).
B_c	Design strength of concrete (kip).
c'	Additional load factor as described in section 1925.2 of the 1994 UBC.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

D	Major thread diameter of threaded anchor or nominal diameter of a bolt (inch).
d_b	Diameter of reinforcing bar (inch).
d_{eff}	Effective diameter used for the calculation of the tensile stress area, A_b .
f'_c	Specified compressive strength of concrete, but not to exceed 6,000 psi (pounds per square inch) for design herein.
f'_s or f_{ut}	Minimum specified tensile strength of anchor steel (psi).
f_y	Minimum specified yield strength of steel (psi).
h	Overall thickness of concrete member (inches).
L_d	Embedment length of an anchor measured from the bearing surface to the top of rough concrete (inches).
l_d	Minimum development length of reinforcement (inches).
m	Edge distance measured from the anchor axis to the free edge (inches).
n	Number of threads per inch.
P'	Ultimate strength of an anchor in tension (kip).
P_c	Design tension strength of concrete (kip).
P_n	Design tension strength of steel anchor (kip).
P_s	Actual working tensile force on the anchor(s) (kip).
P_u	Factored tension loading (kip).
V'	Ultimate strength of an anchor in shear (kip).
V_c	Design strength of concrete (kip).
V_n	Design shear strength of steel anchor (kip).
V_s	Actual working shear force on the anchor(s) (kip).

FLUOR

ANCHOR BOLT DESIGN CRITERIA

V_u	Factored shear loading (kip).
λ	Concrete weight correction factor. Equals 1 for normal weight.
ϕ	Strength reduction factor.
μ	Coefficient of friction. Refer to Attachment 05.
ϕF_t	Allowable tensile stress for steel anchors (ksi).
ϕF_v	Allowable shear stress for steel anchors (ksi).

DEFINITIONS

Anchor Head

A nut, washer, plate, bolt head, or other component designed to transmit anchor loads to the concrete by bearing.

Attachment

Structural components external to the embedment that transmit load to the embedment.

Embedment

The portion of the anchorage system, steel anchors embedded in concrete, or grout designed to transmit loading from the attachment into the concrete. The embedment may be fabricated of plates, shapes, bolts, reinforcing bars, shear connectors, expansion anchors, inserts, or any combination thereof.

Ductile Design

Design of anchorage systems such that in the event of overload the steel anchors will fail before concrete failure occurs. Concrete stress cones should be designed to withstand the ultimate strength of the anchor in tension and shear.

Nonductile Design

Design in which concrete brittle failure may occur at extreme overload. Concrete stress cones will be designed to resist factored design loads rather than ultimate bolt capacities.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

DUCTILE DESIGN PROCEDURE AS DESCRIBED BY ACI

Design Loads

Prior to design of the attachment, transmitted loads will be factored in accordance with ACI applicable codes.

- $P_u = (\text{ACI load factors}) \times P_s$
- $V_u = (\text{ACI load factors}) \times V_s$

Shear forces may be resisted by friction and need not be considered provided the following:

- Nonseismic shear
 - Frictional resistance due to vertical forces and friction resistance due to compression caused by moment couple forces must be greater than the design shear force or bolts and embedments must be capable of carrying the entire applicable shear forces.
- Seismic shear forces
 - Only frictional resistance due to compression as a result of moment couple forces may be used as resistance against seismic shear forces. Frictional resistance must be greater than the factored design loads, or bolts and embedment must be capable of carrying the entire applicable shear force.
 - Current Fluor Daniel practice is to include frictional resistance, as described above, only for vertical vessel anchorage design. Seismic friction resistance is commonly excluded for other cases such as steel columns.

Design of Steel Material Properties

Standard headed anchor bolts or threaded rods with heavy hex nuts will be used. Normally, ASTM (American Society for Testing and Materials) A307 bolts or ASTM A36 threaded rods with ASTM A563 heavy hex nuts, tack welded to the rod to prevent movement, will be specified. Other materials may be used as required.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Steel	f_y (ksi)	f_{ut} (ksi)	Diameter (inches)
A307	36	60	1/4 to 4
A36	36	* 58	1/4 to 8
A193 (Grade B7)	105	125	1/4 to 2-1/2
	95	115	2-5/8 to 4
	75	100	4-1/8 to 7
A449 (Type 1)	92	120	1/4 to 1
	81	105	1-1/8 to 1-1/2
	58	90	1-5/8 to 3

* For design purposes, assume $f_{ut} = 60$ ksi

Design Of Steel Anchors

Steel anchors will be checked for tension, shear, and combined action.

Bolt Tension

Bolt tension will be checked as follows:

$$\begin{aligned} P_u &< \phi P_n \\ \phi P_n &= \phi F_t \times A_b \end{aligned}$$

ϕF_t = the lesser of $0.9f_y$ or $0.8 f_{ut}$

For A36 or A307 = 32.4 or 48 respectively, hence $\phi F_t = 32.4$ ksi

For A_b , refer to Attachment 01.

Bolt Shear

Bolt shear will be checked as follows:

$$\begin{aligned} V_u &< \phi V_n \\ \phi V_n &= \phi F_v A_b \\ \phi F_v &= \text{(by shear friction method)} \\ &= \phi f_y \mu \\ &\text{where } \phi = 0.85 \text{ and } \mu = 0.55 \end{aligned}$$

For A_b , refer to Attachment 01.

Combined Action

Interaction of bolts under tension and shear will be as follows:

FLUOR

ANCHOR BOLT DESIGN CRITERIA

$$\frac{P_u}{\phi P_n} + \frac{V_u}{\phi V_n} \leq 1.0$$

The interaction equation must always be less than or equal to 1.0. ACI load factors have already accounted for wind and seismic short term loading within the equation.

Design Of Embedment

ACI requires that concrete embedments be designed to be ductile. Therefore, the embedments must be designed to withstand the ultimate capacities of the bolt in tension and shear.

Ultimate bolt capacities will be determined as follows:

$$P' = f_{ut} A_b$$

$$V' = \mu f_{ut} A_b$$

Note!!! Where concrete embedment is designed for ultimate bolt capacity, A_b should be effective stress area without corrosion allowance; for example, A_b values listed in Attachment 01, sheet 2, should always be used.

Alternatively, except for the design of nuclear facilities, P' and V' may be taken as 4/3 times the factored design loads.

The requirements herein must be attained:

Equation 1: Tension $P' \leq \phi P_c$; or reinforce appropriately.

Equation 2: Shear $V' < \phi V_c$; or reinforce appropriately.

Requirement 3: Edge distance will never be less than the greater of 4D or 4 inches.

Requirement 4: Where the alternate nonductile design approach is used, the following equation must be satisfied in lieu of Equations 1 and 2, or reinforce appropriately:

$$\left(\frac{P'}{\phi P_c} \right)^2 + \left(\frac{V'}{\phi V_c} \right)^2 \leq 1.0$$

If reinforcement is provided for either tension or shear embedment, that component is removed from the above equation, leaving the equivalent of Equation 1 or 2.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Note!!! Requirement 4 is an empirical interaction equation analogous to that in UBC 1994, Section 1925.3.

GENERAL

Design strength of the concrete embedment is controlled by the following failure modes: tension pullout or lateral bursting, or shear spalling. Strength for each mode is based on an assumed failure surface which propagates at an angle of 45 degrees from the point of application toward the concrete surface. Equations have been developed based on a uniform tensile stress of $4\sqrt{f'_c}$ acting on an effective stress area. Refer to ACI 349R B.4.2. The stress area is defined by the projected area of stress cones which radiate toward the concrete face. The effective area will be limited by overlapping stress cones, intersection of cones with concrete edges, the bearing area of an anchor head, and by the overall thickness of the concrete. Refer to Attachment 04 for the determination of the effective stress area, A_p .

Strength Reduction Factors

- ϕ = 0.85 for embedments anchored beyond the member far face reinforcement.
- ϕ = 0.85 for embedments anchored in the compression zone of a member.
- ϕ = 0.85 where embedments are in the tension zone but tensile stress of plain concrete based on an uncracked section is less than $0.65 \times 5\sqrt{f'_c}$.
- ϕ = 0.65 for all other embedments.

Tension Pullout Design Strength

The concrete failure cone will propagate from the bearing edge of the anchor head as shown in Attachment 05, Figure 2a. Reductions of strength accounting for geometric layout will be considered in the determination of the effective stress area, A_p .

Concrete tension capacity is proportional to the anchor bolt length. The lengths of bolts shown in Structural Engineering Practice 670.215.4050: Standard Anchor Bolts and Sleeves - Design Details, have been provided as a guide and to provide consistency throughout projects. The lengths have been based on the length necessary to develop tension reinforcement where it is required. Other lengths may be used where necessary, provided the requirements herein are maintained.

Cone Pullout Design Capacity will be determined as follows:

FLUOR

ANCHOR BOLT DESIGN CRITERIA

$$\phi P_c = \phi 4 \sqrt{f'_c} A_p$$

Where concrete strength does not meet the requirements of Equation 1, reinforcement must be provided. Refer to Attachment 05, Figure 2b, for details.

$$A_{st} = \frac{P'}{0.9 f_y}$$

Reinforcement will be oriented in a manner that restricts propagation of cracking should it occur. To accomplish this, reinforcement must be fully developed on both sides of the assumed failure surface. It is recommended that reinforcement be placed concentric with the failure cone. In addition, reinforcement will not be placed farther than $L_d/3$ from the axis of the anchor.

Note!!! Typical pier reinforcement may be considered as tensile resisting elements according to the above criteria, provided the bars can develop adequate length within the free side of the failure cone.

Lateral Bursting Design Strength

The minimum edge distance at which the cone has sufficient strength according to ductile design methods has been determined to be $3.6D$. Refer to ACI 349R B.5.1.1. This practice limits the edge distance to $4D$ or 4 inches, hence lateral bursting need not be addressed unless an unusual situation occurs.

When an anchor subject to tensile force is located closer than $3.6D$, lateral bursting failure may occur rather than tension pullout. This is due to differences in the restraint stiffness around the periphery of the anchor head which tends to cause lateral strain concentration on the side of the free edge. This concentration will cause a blowout cone failure that propagates from the anchor head toward the free edge as shown in Attachment 10, Figure 1.

Ultimate Bursting Design Load

$$B' = \beta \times P'$$

where

$$\beta = 0.25$$

as a result of Poisson's effect.

Concrete Strength

Concrete strength will be greater than the ultimate design load.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

$$\phi B_c = \phi 4 \sqrt{f'_c} A_p$$

Reinforcement

When reinforcement is required, it will be placed in a similar manner to reinforcement for shear spalling failure. The area required will be as follows:

$$A_{sb} = \frac{B'}{(0.9f_y)}$$

Shear Spalling Design Strength

The concrete failure cone will propagate from the bolt bearing at the surface of the concrete toward the loaded edge as shown in Attachment 06, Figure 1. Strength is determined on the same premise as the tension failure mode with the exception that only half the stress cone is available to provide resistance.

Shear spalling design capacity will be determined as follows:

$$\phi V_c = \phi 4 \sqrt{f'_c} A_p$$

Note that for ductile design (and $f'_c = 4,000$ psi and 36 ksi bolt material), Requirement 2 will be satisfied for edge distances of 10D or greater. This assumes that the shear cone is not reduced due to adjacent bolts or pedestal dimensions.

Where concrete strength does not meet the requirements of Equation 2 (or Requirement 4 as applicable), reinforcement will be provided. For details, refer to Attachment 06, Figure 2.

$$A_{sv} = \frac{V'}{(0.9f_y)}$$

Reinforcement will be oriented in a manner that restricts cracking should it occur. Several approaches have been taken to provide adequate reinforcement. Development length for any size rebar on the free side of the assumed crack is nearly impossible, because edge distances that require reinforcement are generally less than the development length of a No. 4 bar. Current Fluor Daniel practice is to provide reinforcing ties that penetrate concentrically through the assumed failure cone. Other details may be used where the engineer can demonstrate physical adequacy within economical limits.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

REFERENCES

ACI (American Concrete Institute)

ACI Building Code Requirements for Reinforced Concrete, (ACI 318-95):
Chapter 11, 1995.

ACI Code Requirements for Nuclear Safety Related Concrete Structures, (ACI
349-85): Appendix B, 1985.

AISC (American Institute of Steel Construction Inc.), Manual of Steel Construction,
ASD 9th Ed., USA, Section 4, 1989.

ASTM (American Society for Testing and Materials)

A36

A307

A563

ICBO (International Conference of Building Officials), UBC (Uniform Building Code),
1994Ed., Section 1925.

Ministry of Energy, Seismic Design of Petrochemical Plants, Vol. 1, New Zealand, 1981.

Structural Engineering

Practice 670.215.4050: Standard Anchor Bolts And Sleeves - Design Details

ATTACHMENTS

Attachment 01:

Bolt Design Strength, ACI 349 Appendix B (A36 And A307 Steel Anchors)

Attachment 02:

UBC Design Procedure For Headed Type Anchors

Attachment 03:

Bolt Design Strength, UBC 1925 (A36 And A307 Steel Anchors)

Attachment 04:

Effective Stress Area Of Concrete

Attachment 05:

Figure 1: Coefficients Of Friction

Figure 2: Concrete Pullout Failure

ANCHOR BOLT DESIGN CRITERIA**Attachment 06:**

Concrete Shear Spalling Failure

Figure 1.

Figure 2.

Attachment 07:

Figure 1: Overlapping Stress Cones

Figure 2: Enlargement Of Stress Area At Overlapping Stress Cones Or Edge Intersection

Attachment 08:

Multiple Cone Overlapping

Attachment 09:

Figure 1: Two-Way Shear Failure

Figure 2: Vertical Vessel Reinforcement Detail

Attachment 10:

Figure 1: Lateral Bursting Failure

Figure 2: Yielding Mechanism For All Tubular Equipment

Attachment 11:

Special Design Consideration For Tall Tubular Equipment

Attachment 12:

Use Of Sleeves

Attachment 13:

Sample Design 1: Steel Column On A Concrete Pier

Attachment 14:

Sample Design 2: Steel Column On A Concrete Pier

Attachment 15:

Sample Design 3: Horizontal Exchanger On A Concrete Pier

Attachment 16:

Sample Design 4: Vertical Vessel On A Concrete Pier

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Bolt Design Strength, ACI349 Appendix B(A36 and A307 Steel Anchors)

Nongalvanized							
D(in)	d _{eff}	n	A _b	øP _n	øV _n	A _{st}	A _{sv}
1/2	0.36	13.00	0.10	3.24	1.68	0.11	0.06
5/8	0.47	11.00	0.17	5.51	2.86	0.19	0.10
3/4	0.53	10.00	0.22	7.13	3.70	0.24	0.13
7/8	0.64	9.00	0.32	10.37	5.39	0.36	0.20
1.00	0.75	8.00	0.44	14.26	7.41	0.49	0.27
1-1/4	0.99	7.00	0.77	24.95	12.96	0.86	0.47
1-1/2	1.21	6.00	1.15	37.26	19.35	1.28	0.70
1-3/4	1.43	5.00	1.61	52.16	27.10	1.79	0.98
2.00	1.66	4.50	2.16	69.98	36.35	2.40	1.32
2-1/4	1.91	4.50	2.87	92.99	48.30	3.19	1.75
2-1/2	2.13	4.00	3.56	115.34	59.91	3.96	2.18
2-3/4	2.38	4.00	4.45	144.18	74.89	4.94	2.72
3.00	2.63	4.00	5.43	175.93	91.39	6.03	3.32

- Note!!!**
1. D = Nominal bolt diameter
 2. d_{eff} = Effective bolt diameter with an additional corrosion reduction of 1/16" for 5/8-inch bolts and smaller or 1/8" for larger bolts.
 3. A_b = Effective stress area
A_b = $\pi/4 \times d_{eff}^2$
 4. øP_n = øF_tA_b : øF_t = lesser of 0.9f_y or 0.8f_{ut}
 5. øV_n = øF_vA_b : øF_v = 0.55 x 0.85f_y
 6. A_{st} = (f_{ut}A_b) / (0.9f_y (rebar))
 7. A_{sv} = (f_{ut}A_bμ) / (0.9f_y (rebar))

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Bolt Design Strength, ACI349 Appendix B(A36 and A307 Steel Anchors)

Galvanized							
D(in)	d _{eff}	n	A _b	øP _n	øV _n	A _{st}	A _{sv}
1/2	0.43	13.00	0.15	4.86	2.52	0.17	0.09
5/8	0.54	11.00	0.23	7.45	3.87	0.26	0.14
3/4	0.65	10.00	0.33	10.69	5.55	0.37	0.20
7/8	0.77	9.00	0.47	15.23	7.91	0.52	0.29
1.00	0.88	8.00	0.61	19.76	10.27	0.68	0.37
1-1/4	1.11	7.00	0.97	31.43	16.33	1.08	0.59
1-1/2	1.34	6.00	1.41	45.68	23.73	1.57	0.86
1-3/4	1.56	5.00	1.91	61.88	32.15	2.12	1.17
2.00	1.78	4.50	2.49	80.68	41.91	2.77	1.52
2-1/4	2.03	4.50	3.24	104.98	54.53	3.60	1.98
2-1/2	2.26	4.00	4.01	129.92	67.49	4.46	2.45
2-3/4	2.51	4.00	4.95	160.38	83.31	5.50	3.03
3.00	2.76	4.00	5.98	193.75	100.64	6.64	3.65

Note!!!	1.	D	=	Nominal bolt diameter			
	2.	d _{eff}	=	Effective bolt diameter with an additional corrosion reduction of 1/16" for 5/8-inch bolts and smaller or 1/8" for larger bolts.			
	3.	A _b	=	Effective stress area			
		A _b	=	$\pi/4 \times d_{eff}^2$			
	4.	øP _n	=	øF _t A _b :	øF _t	=	lesser of 0.9f _y or 0.8f _{ut}
	5.	øV _n	=	øF _v A _b :	øF _v	=	0.55 x 0.85f _y
	6.	A _{st}	=	(f _{ut} A _b) / (0.9f _y (rebar))			
	7.	A _{sv}	=	(f _{ut} A _b x μ) / (0.9f _y (rebar))			

FLUOR

ANCHOR BOLT DESIGN CRITERIA

UBC Design Procedure for Headed-Type Anchors

General

The UBC method has the same premise as the ACI method but it is not as stringent as the ACI concerning ductile design. The equations are slightly different from the ACI and should not be used interchangeably. ACI methods are preferred for designs in the UBC defined seismic zones 3 and 4.

- Load Factors
Prior to the design of the attachment, loads will be factored in accordance with UBC Section 1909.2 and UBC Section 1925.2.

$$P_u = (\text{factors of 1909.2} \times 1925.2) \times P_s$$

$$V_u = (\text{factors of 1909.2} \times 1925.2) \times V_s$$

Load Factors of UBC 1925.2 are as follows:

$$c' = 2 \text{ for cases without special inspection.}$$

$$c' = 1.3 \text{ for cases where special inspection is provided.}$$

$$c' = 3 \text{ for anchors in the tension zone without special inspection.}$$

$$c' = 2 \text{ for anchors in the tension zone where special inspection is provided.}$$

- Shear forces may be resisted by friction and need not be considered provided the following:

Nonseismic Shear

Frictional resistance due to vertical forces and friction resistance due to compression caused by moment couple forces must be greater than the factored design shear force or bolts and embedments must be capable of carrying the entire applicable shear forces.

Seismic Shear Forces

Only frictional resistance due to compression as a result of moment couple forces may be used as resistance against seismic shear forces. Frictional resistance must be greater than the factored design loads, or bolts and embedment must be capable of carrying the entire applicable shear force.

Current Fluor Daniel practice is to include frictional resistance, as described above, only for vertical vessel anchorage design. Seismic friction resistance is commonly excluded for other cases such as steel columns.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

UBC Design Procedure for Headed-Type Anchors

Design of Steel Anchors

Steel anchors will be checked for tension, shear, and combined action.

- Bolt Tension: Bolt tension will be checked as follows:

$$\begin{aligned} P_u &< \phi P_n \\ \phi P_n &= \phi F_t \times A_b \\ \phi F_t &= 0.9 f_{ut} \end{aligned}$$

For A_b , refer to Attachment 03.

- Bolt Shear: Bolt shear will be checked as follows:

$$\begin{aligned} V_u &< \phi V_n \\ \phi V_n &= \phi F_v \times A_b \\ \phi F_v &= 0.75 f_{ut}, \text{ for A36 and A307} = 45 \text{ ksi} \end{aligned}$$

For A_b , refer to Attachment 03.

- Combined Action: Combined tension and shear must satisfy the following interaction equation.

$$\left(\frac{P_u}{\phi P_n} \right)^2 + \left(\frac{V_u}{\phi V_n} \right)^2 < 1.0$$

Note!!! The above equation must always be less than or equal to 1.0. Load factors have already accounted for short term loading situations.

Design of Embedment

- Code Requirements

The following requirements must be attained:

- Equation 1: Tension $P_u < \phi P_c$; or reinforce appropriately.
- Equation 2: Shear $V_u < \phi V_c$; or reinforce appropriately.
- Equation 3: Combined action $\frac{1}{\phi} \left[\left(\frac{P_u}{P_c} \right)^2 + \left(\frac{V_u}{V_c} \right)^2 \right] < 1.0$
- Requirement 4: Edge Distance will never be less than the greater of 4D or 4in.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

UBC Design Procedure for Headed-Type Anchors

- **General**

Design strength of the concrete embedment is controlled by the following failure modes, tension pullout or shear spalling. Strength for each mode is based on an assumed failure surface which propagates at an angle of 45 degrees from the point of application toward the concrete surface. Equations have been developed based on a uniform tensile stress of $4\sqrt{f'_c}$ acting on an effective stress area. The stress area is defined by the projected area of stress cones which radiate toward the concrete face. The effective area will be limited by overlapping stress cones, intersection of cones with concrete edges, the bearing area of an anchor head, and by the overall thickness of the concrete. Refer to Attachment 04 for the determination of the effective stress area, A_p .

Strength Reduction Factors:

$$\phi = 0.65$$

- **Tension Pullout Design Strength**

The concrete failure cone will propagate from the bearing edge of the anchor head as shown in Attachment 05, Figure 2a. Reductions of strength accounting for geometric layout will be considered in the determination of the effective stress area, A_p .

Concrete tension capacity is proportional to the anchor bolt length. The lengths of bolts shown in Practice 670.215.4050: Standard Anchor Bolts and Sleeves - Design Details, have been provided as a guide and to provide consistency throughout projects. The lengths have been based on the length necessary to develop tension reinforcement where it is required. Other lengths may be used as necessary, when the requirements herein are maintained.

Cone pullout design capacity will be determined as follows:

$$\phi P_c = \phi \lambda 4 \sqrt{f'_c} A_p$$

Wherever concrete strength does not meet the requirements of Equation 1, reinforcement must be provided. Refer to Attachment 05, Figure 2b for details.

$$A_{st} = \frac{P_u}{(0.9f_y)}$$

Reinforcement will be oriented in a manner that restricts propagation of cracking should it occur. To accomplish this, reinforcement must be fully developed on both sides of the assumed failure surface. It is recommended that reinforcement be placed concentric with the failure cone. In addition, reinforcement will not be placed farther than $8d_b$ or $L_d/3$ from the axis of the anchor.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

UBC Design Procedure for Headed-Type Anchors

Note!!! Typical pier reinforcement may be considered as tensile resisting elements according to the above criteria, provided the bars can develop adequate length within the free side of the failure cone.

- **Shear Spalling Design Strength**

The concrete failure cone will propagate from the bolt bearing at the surface of the concrete toward the loaded edge as shown in Attachment 06. Strength is determined on the same premise as the tension failure mode with the exception that only half the stress cone is available to provide resistance.

Shear spalling design capacity will be determined as follows:

Where edge distance > 10D,

$$\phi V_c = \phi \lambda 4 \sqrt{f'_c} (200 \times A_b)$$

Satisfy Equation 2 or reinforce appropriately.

Where 4D < edge distance < 10D,

$$\phi V_c = \phi \lambda 4 \sqrt{f'_c} A_b$$

Satisfy Equation 2 or reinforce appropriately. (If Equation 2 is not satisfied, increasing the edge distance is strongly recommended.)

Edge distance will not be less than 4D or 4 inches.

Where concrete strength does not meet the requirements of Equation 2 or edge distance is less than 10D, reinforcement will be provided. For details refer to Attachment 06.

$$A_{sv} = \frac{V_u}{(0.9 f_y)}$$

Reinforcement will be oriented in a manner that restricts cracking. Several approaches have been taken to provide adequate reinforcement. However, developing any size rebar on the free side of the assumed crack is nearly impossible, because edge distances that require reinforcement are generally less than the development length of even the smallest of rebar sizes. This is the basis for recommending increasing the edge distance when Equation 2 is not satisfied. Where it is impractical to increase the edge distance to satisfy Equation 2, current Fluor Daniel practice is to provide reinforcing ties that penetrate concentrically through the assumed failure cone.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Bolt Design Strength, UBC 1925 (A36 and A307 Steel Anchors)

D(in)	d _{eff}	Nongalvanized			
		n	A _b	øP _n	øV _n
1/2	0.36	13.00	0.10	5.40	4.50
5/8	0.47	11.00	0.17	9.18	7.65
3/4	0.53	10.00	0.22	11.88	9.90
7/8	0.64	9.00	0.32	17.28	14.40
1.00	0.75	8.00	0.44	23.76	19.80
1-1/4	0.99	7.00	0.77	41.58	34.65
1-1/2	1.21	6.00	1.15	62.10	51.75
1-3/4	1.43	5.00	1.61	86.94	72.45
2.00	1.66	4.50	2.16	116.64	97.20
2-1/4	1.91	4.50	2.87	154.98	129.15
2-1/2	2.13	4.00	3.56	192.24	160.20
2-3/4	2.38	4.00	4.45	240.30	200.25
3.00	2.63	4.00	5.43	293.22	244.35

- Note!!!**
1. D = Nominal bolt diameter
 2. d_{eff} = Effective bolt diameter
 3. A_b = Effective stress area

$$A_b = \frac{\pi}{4} d_{eff}^2$$
 4. øP_n = øF_tA_b, øF_t = 0.9f'_s
 5. øV_n = øF_vA_b, øF_v = 0.75f'_s

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ANCHOR BOLT DESIGN CRITERIA

Bolt Design Strength, UBC 1925 (A36 and A307 Steel Anchors)

Galvanized					
D(in)	d _{eff}	n	A _b	øP _n	øV _n
1/2	0.43	13.00	0.15	8.10	6.75
5/8	0.54	11.00	0.23	12.42	10.35
3/4	0.65	10.00	0.33	17.82	14.85
7/8	0.77	9.00	0.47	25.38	21.15
1.00	0.88	8.00	0.61	32.94	27.45
1-1/4	1.11	7.00	0.97	52.38	43.65
1-1/2	1.34	6.00	1.41	76.14	63.45
1-3/4	1.56	5.00	1.91	103.14	85.95
2.00	1.78	4.50	2.49	134.46	112.05
2-1/4	2.03	4.50	3.24	174.96	145.80
2-1/2	2.26	4.00	4.01	216.54	180.45
2-3/4	2.51	4.00	4.95	267.30	222.75
3.00	2.76	4.00	5.98	322.92	269.10

- Note!!!**
1. D = Nominal bolt diameter
 2. d_{eff} = Effective bolt diameter
 3. A_b = Effective stress area

$$A_b = \frac{\pi}{4} d_{eff}^2$$
 4. øP_n = øF_tA_b, øF_t = 0.9f'_s
 5. øV_n = øF_vA_b, øF_v = 0.75f'_s

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ANCHOR BOLT DESIGN CRITERIA

Effective Stress Area of Concrete

General

This attachment has been provided as an aid for calculation and should be interpreted neither as standard nor as code. Current codes do not provide in depth detail on the procedure for calculating the effective stress area. Several assumptions have been left for the engineer's judgment of particular situations.

- Because failure is initiated at the periphery of the anchor head, the area of the head itself does not contribute to resistive strength and should be subtracted for all computations of A_p . (Refer to Attachment 05, Figure 2.)
- For overlapping stress cones or intersection with an edge, refer to Attachment 07, Figures 1 and 2.
- Calculation for multiple stress cones (refer to Attachment 08, Figure 1) where $e < 0.707 r$. Where $e > 0.707 r$ (refer to Attachment 07, Figures 1 and 2).
- When the overall concrete dimension is small (anchorage to slabs or walls), the effect of 2-way shear must be considered. Reduction of effective stress area will be in accordance with ACI 349, Appendix B. Refer to Attachment 09, Figure 1.

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ANCHOR BOLT DESIGN CRITERIA

Figure 1& 2: Coefficients of Friction and Concrete Pullout Failure Respectively

Figure 1. Coefficients of Friction

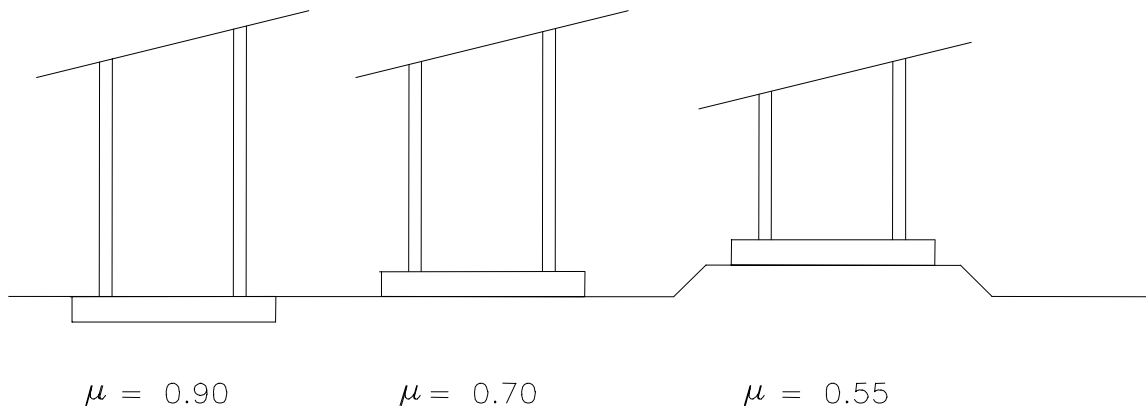
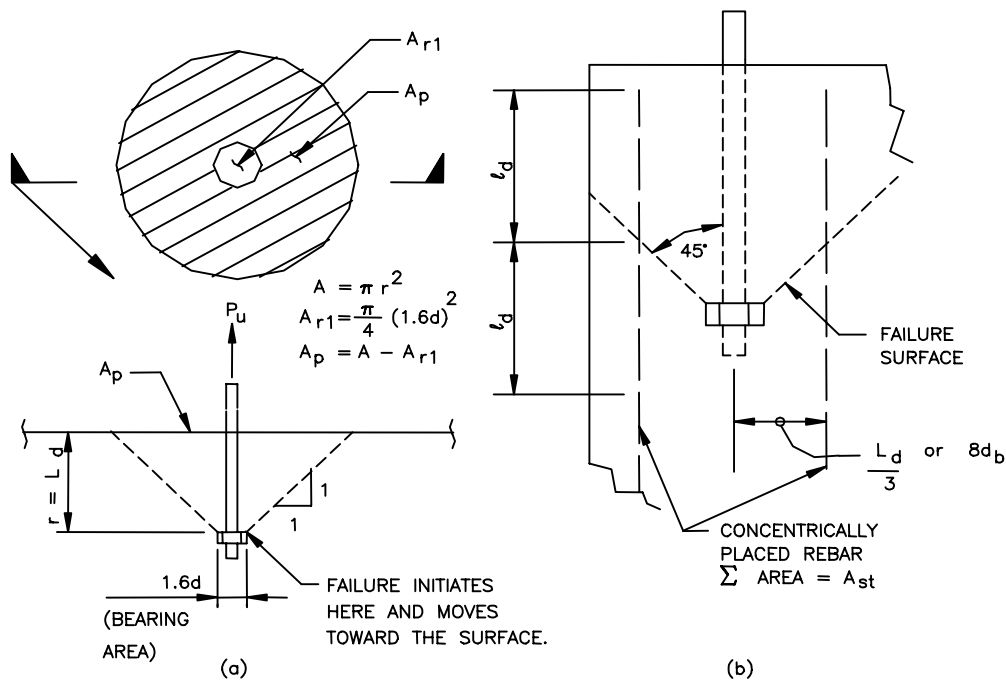


Figure 2. Concrete Pullout Failure



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ANCHOR BOLT DESIGN CRITERIA

Concrete Shear Spalling Failure

Figure 1

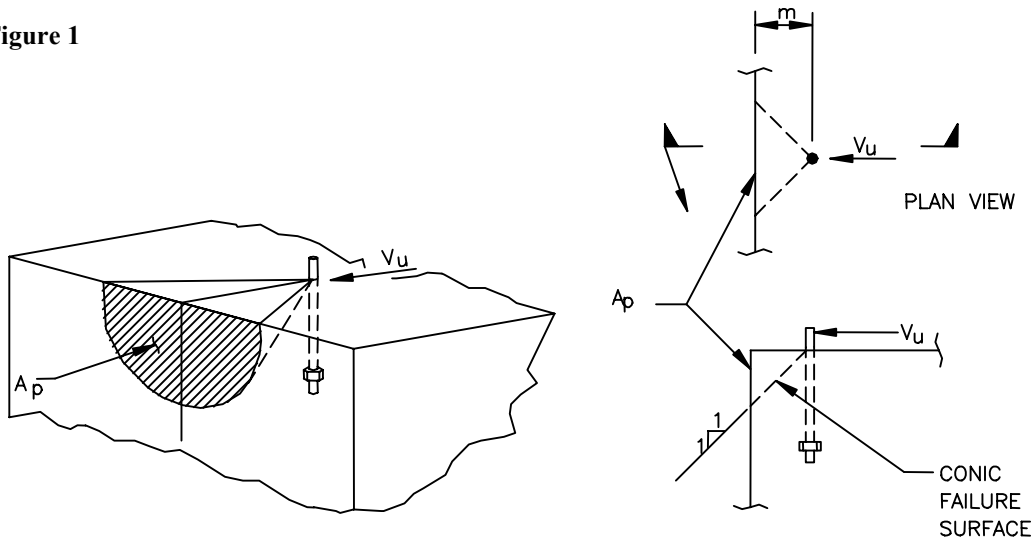
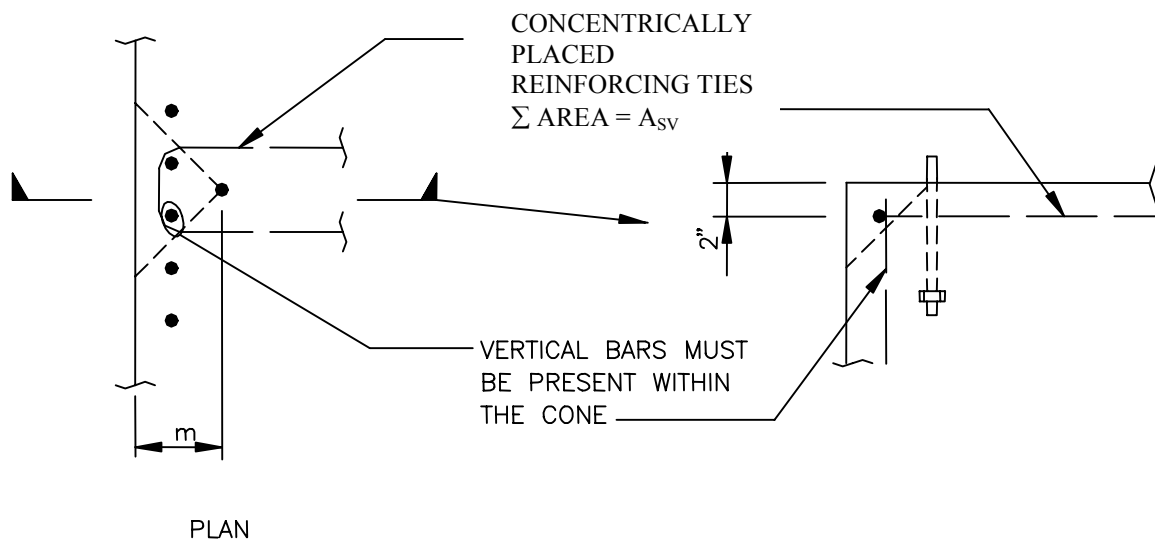


Figure 2



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ANCHOR BOLT DESIGN CRITERIA

Figure 1 & 2 : Overlapping Stress Cones and Enlargement of Stress Area at Overlapping Stress Cones or Edge Intersection

Figure 1. Overlapping Stress Cones

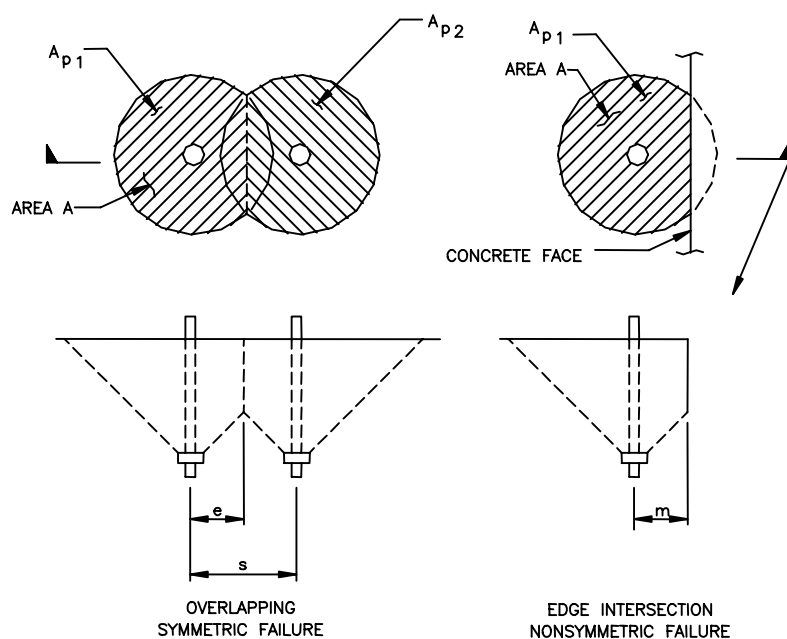
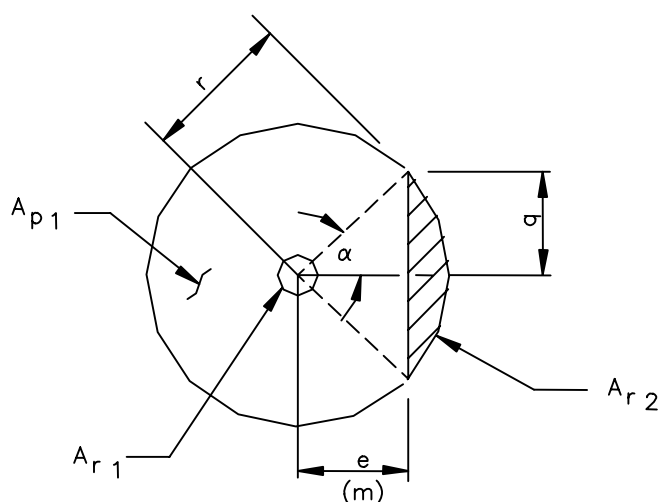


Figure 2. Enlargement of Stress Area at Overlapping Stress Cones or Edge Intersection



$$\alpha = \cos^{-1} \left(\frac{e}{r} \right)$$

$$q = r \sin \alpha$$

$$A = \pi r^2$$

$$A_{r1} = \frac{\pi}{4} (1.6D)^2$$

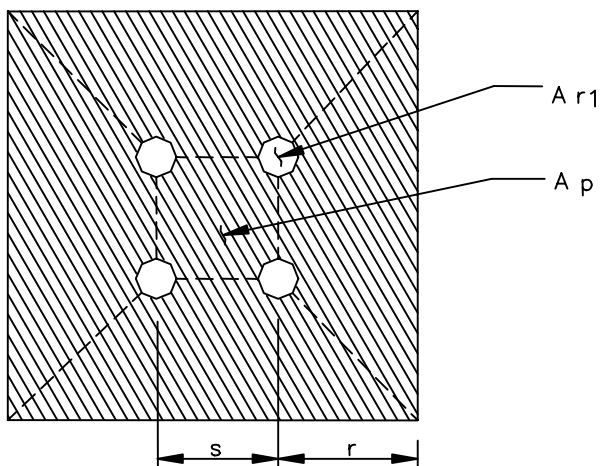
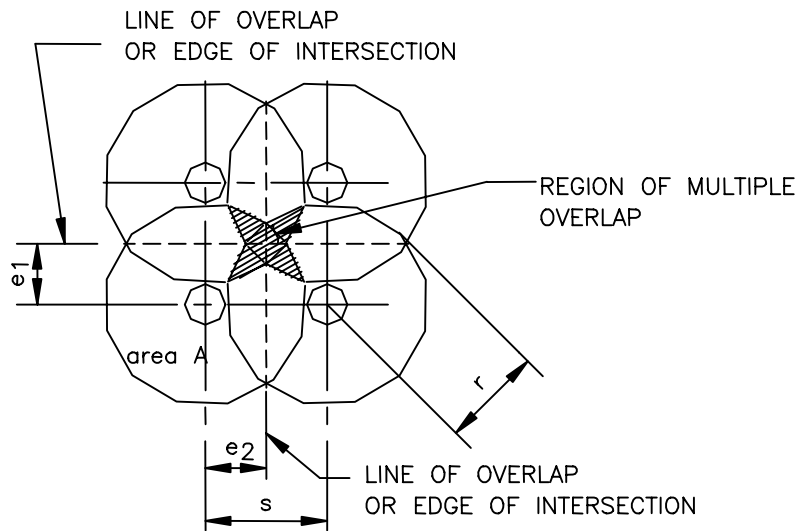
$$A_{r2} = \frac{\pi r^2 \alpha}{180} - qe$$

$$A_p = A - A_{r1} - A_{r2}$$

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ANCHOR BOLT DESIGN CRITERIA

Multiple Cone Overlapping



A PYRAMID TYPE FAILURE MAY BE ASSUMED:

1.) OVERLAPPING

$$A_p = (s + 2r)^2 - 4 A r_1$$

2.) SPECIAL CONSIDERATION MUST BE GIVEN TO THE CASE WHERE FAILURE SURFACES INTERSECT WITH CONCRETE FACES.

ANCHOR BOLT DESIGN CRITERIA

Figure 1: Two-Way Shear Failure

Figure 2: Vertical Vessel Reinforcement Details

Figure 1. Two-Way Shear Failure

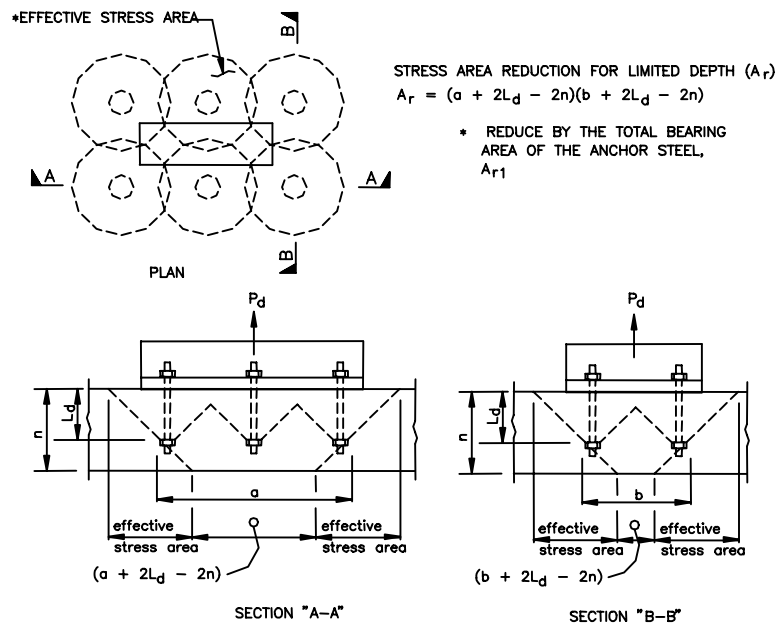
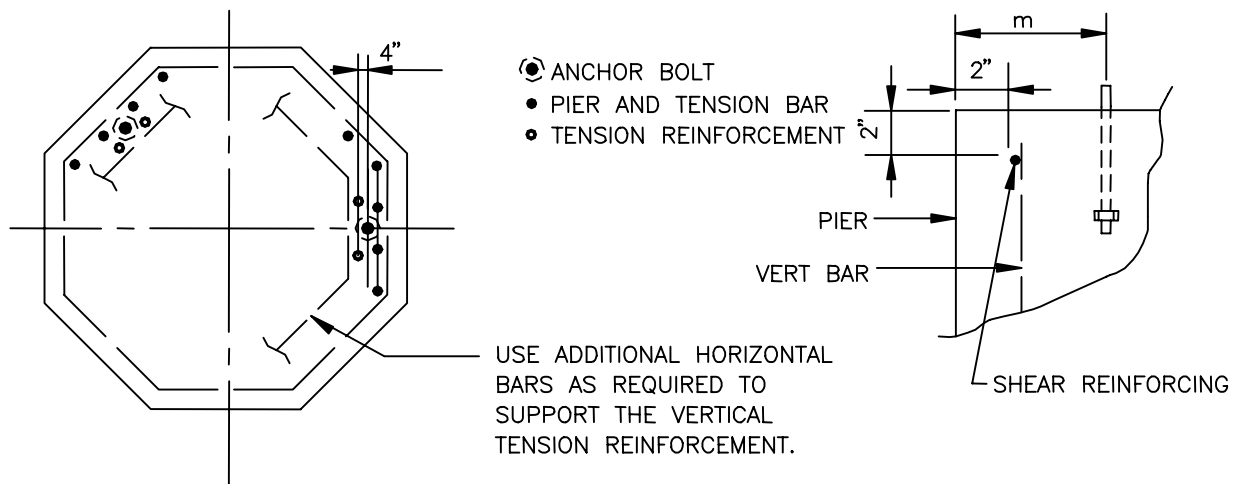


Figure 2. Vertical Vessel Reinforcement Details



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ANCHOR BOLT DESIGN CRITERIA

Figure 1: Lateral Bursting Failure and Figure 2: Yielding Mechanism for All Tubular Equipment

Figure 1. Lateral Bursting Failure

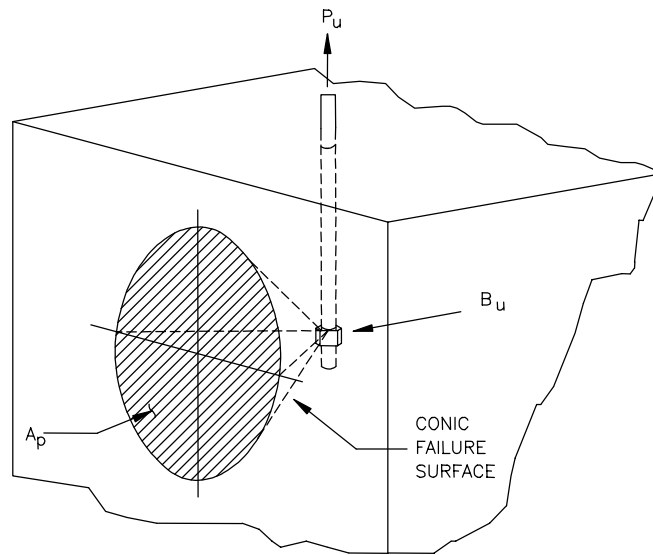
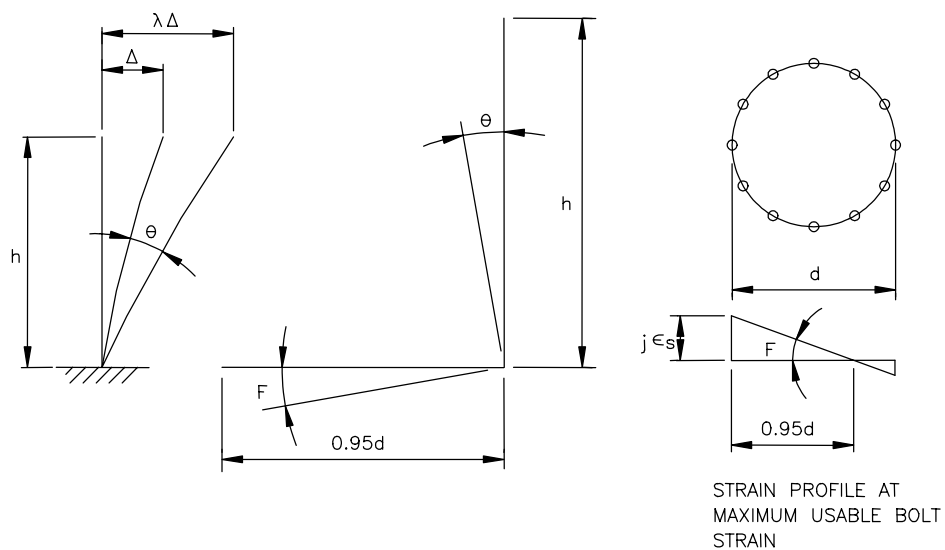


Figure 2. Yielding Mechanism for all Tubular Equipment



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ANCHOR BOLT DESIGN CRITERIA

Special Design Consideration for Tall Tubular Equipment

General

This attachment has been provided primarily for the purpose of informing the structural engineer of a possible problem with the design of tall tubular equipment. Typically, calculations such as these will be the responsibility of the vessel group. However, there may be some instances where it is necessary to design according to the following.

A tall tubular vessel will be defined as a vessel that has a height to diameter ratio greater than 5 and a height greater than 35'- 0".

Problem

The problem is to provide a rational yielding hierarchy of the primary structural system for designs subject to dynamic loading. Dynamic loads include seismic and wind forces.

Ductile behavior may be achieved by designing a ductile yielding mechanism at the base of tall tubular equipment. This can be accomplished by designing a bolt and bolt chair according to the following procedure.

Notation

Δ	=	Elastic displacement of the equipment, having a fixed base. The maximum deflection is limited to $0.01 \times 12 \times h$, in inches.
λ	=	Deformation modification factor = 1.5μ .
μ	=	Ductility factor, taken as 3.
ϵ	=	Maximum usable bolt strain, limited to 0.04.
d	=	Bolt circle diameter, in feet.
h	=	Equipment height, in feet.
j	=	Length of bolt, above top of concrete, for example bolt chair height, required to provide a yielding mechanism, in inches. The maximum for practical purposes will be taken as 18 inches.

Design

$$j = \frac{25d(\lambda - 1)\Delta}{h}$$

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ANCHOR BOLT DESIGN CRITERIA

Special Design Consideration for Tall Tubular Equipment

For derivation, refer to the following example:

$$h = 90' - 0''$$

$$d = 8' - 0'' \quad j = \frac{25 \times 4' \times (4.5 - 1) \times 4''}{90'}$$

$$\Delta = 0' - 4''$$

$$\lambda = 1.5 \times 3 = 4.5 \quad j = 15.56 \text{ inches}$$

Hence, the bolt chair height or bolt length above top of concrete will be 15.56 inches or greater.

Derivation (Refer to Attachment 10, Figure 2)

$$\theta = \frac{(\lambda - 1)\Delta}{h}$$

$$\varphi = \frac{\varepsilon}{0.95d}$$

$$\text{Since } \theta = \varphi j$$

$$\frac{(\lambda - 1)\Delta}{h} = \frac{j\varepsilon}{0.95d}$$

$$\varepsilon = 0.04 \text{ (maximum usable bolt strain)}$$

$$\therefore j = \frac{0.95(\lambda - 1)\Delta}{h\varepsilon}$$

$$j = \frac{25(\lambda - 1)\Delta}{h}$$

j = Bolt Length (from top of concrete to the bottom of nut at the bolt chair)

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ANCHOR BOLT DESIGN CRITERIA

Use of Sleeves

Recommended practice for the use of Sleeves and 2 nuts with standard anchor bolts.
All other applications will receive 1 nut for each bolt

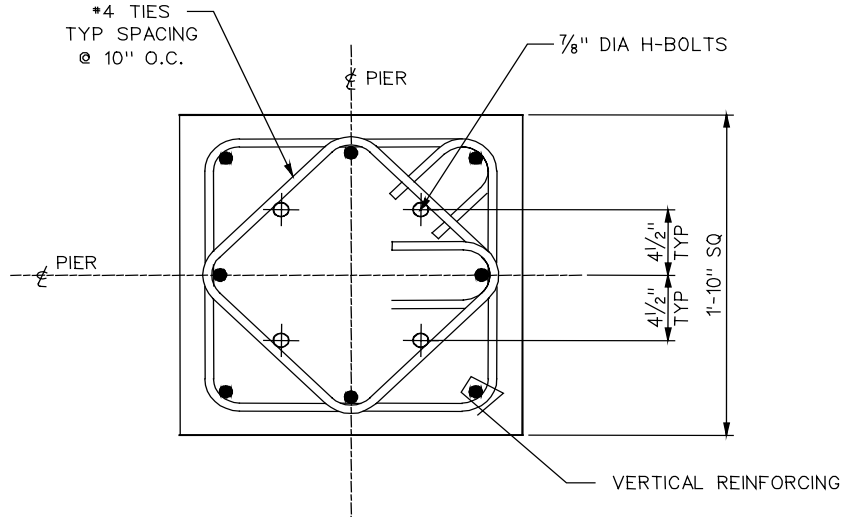
Anchor Bolt Sleeves			
Equipment	Without Sleeves	With Sleeves	With 2 Nuts*
Vertical vessel		X	X
Sphere		X	X
Stack, freestanding		X	X
Structure, major	X		
Pipe support, major	X		
Compressor		X	
Heater, shop assembled		X	
Boiler, shop assembled		X	
Pump		X	
Boiler, field assembled	X		
Heater, field assembled	X		
Pipe support, minor	X		
Horizontal vessel	X		
Exchanger	X		
Stack, guyed	X		
Stack, derrick	X		
Structure, minor	X		
Miscellaneous	X		

* Special Cases

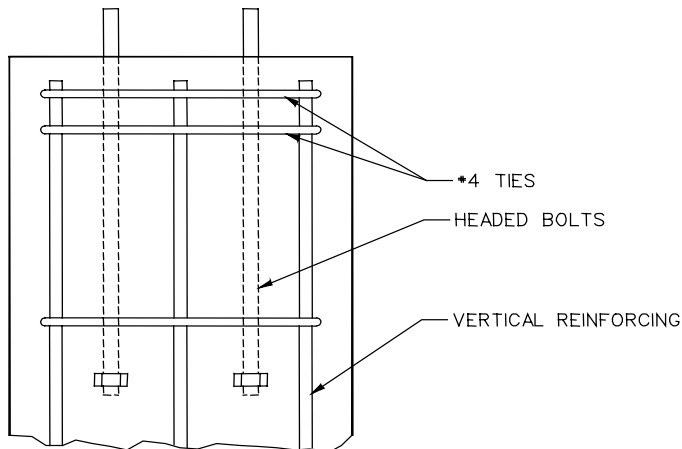
FLUOR

ANCHOR BOLT DESIGN CRITERIA

Sample Design 1: Steel Column on a Concrete Pier



PLAIN
TYPICAL (8) VERTICAL BAR DETAIL



SECTION

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Sample Design 1: Steel Column on a Concrete Pier

Design Data

22" square pier $f_c = 4$ ksi $f_y = 60$ ksi rebar, Standard base plate detail 10 Ref. (670 215 1207)

7/8" ϕ - headed anchors. A307 $f_y = 36$ ksi, $f_{ut} = 60$ ksi

Loads @ Top of Pier

$P = 2$ kip $\downarrow$

$M_s = 8.6$ ft-kip

$V_s = 1$ kip

P_s = Tension force in bolt determined by methods beyond the scope of this practice.

Design

Bolt tension force $P_s = \frac{9.73k}{2\text{bolts}} \uparrow$

Design per ACI-349 method

Since loads are primarily due to seismic action, friction between base plate and concrete is excluded.

$$P_u = 0.75 (1.7 \times 1.1)(P_s) = 13.62 \text{ kip}$$

$$V_u = 0.75 (1.7 \times 1.1)(V_s) = 1.4 \text{ kip}$$

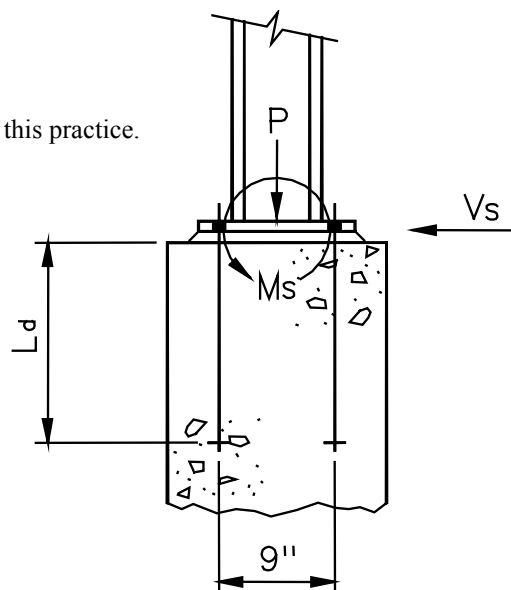
(for non-seismic loads, friction = $\phi(P_s) (N) = (0.85)(9.73)(0.55) = 4.5 \text{ kip} > V_u$, no shear on bolts)

Design of Bolts

Design strength per bolt

$$\begin{aligned} \phi P_n &= 0.9(36 \text{ ksi})(A_b) & A_b &\cong 0.32 \text{ in}^2 \text{ (non-galvanized)} \\ &= 10.37 \text{ kip (Attachment 1)} & &\geq (13.62)/2 = 6.81 \text{ kip} \quad \text{ok} \end{aligned}$$

$$\phi V_c = 0.55(0.85)(36 \text{ ksi})(A_b)$$



ANCHOR BOLT DESIGN CRITERIA

Sample Design 1: Steel Column on a Concrete Pier

$$= 5.39 \text{ kip (Attachment 1)} \geq (1.4)/4 = 0.35 \quad \text{ok}$$

Interaction:

$$\frac{6.81}{10.37} + \frac{0.35}{5.39} = 0.72 \leq 1.0 \quad \text{ok}$$

Design of Concrete

$$L_d = 10", \text{ bolt spacing} = 9", \text{ edge distance} = 6-1/2"$$

$$A_{TOT} = (16-1/2")(22") = 363 \text{ in}^2$$

$$A_{r1} = 1.54 \text{ in}^2$$

Tension:

$$A_p = 363 - 2(1.54)$$

$$= 360 \text{ in}^2$$

Shear:

$$A_p = (282.9)/2 = 141.5 \text{ in}^2$$

Concrete Pyramid Strength

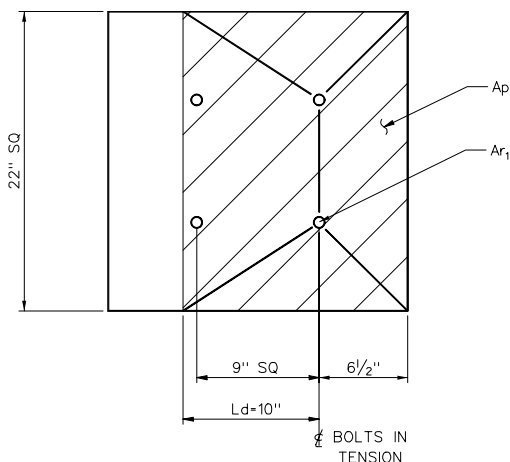
$$\phi P_c = \phi (4) \sqrt{f'_c} (A_p)$$

$$= 0.65(4) \frac{\sqrt{4000}}{1000} (360) = 59.2 \text{ kip per 2 bolts}$$

$$P' = f_{ut} (A_b^*) = 60 \text{ ksi}(0.47 \text{ in}^2) = 28.2 \text{ kip per bolt}$$

$$= 56.4 \text{ kip per 2 bolts}$$

$$\phi P_c \geq P' \quad \text{ok} \quad * A_b \text{ is based on uncorroded, or galvanized bolt diameter.}$$



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ANCHOR BOLT DESIGN CRITERIA

Sample Design 1: Steel Column on a Concrete Pier

$$\phi V_c = 0.65(4) \frac{\sqrt{4000}}{1000} (141.5) = 23.27 \text{ kip per 2 bolts}$$

$$V' = 0.55(60 \text{ ksi})(0.47 \text{ in}^2) = 15.5 \text{ kip per bolt}$$

$$= 31.0 \text{ kip per 2 bolts}$$

$$\phi V_c < V' \quad \text{edge} = 6\text{-}1/2" < (10\text{-}7/8") = 5.75$$

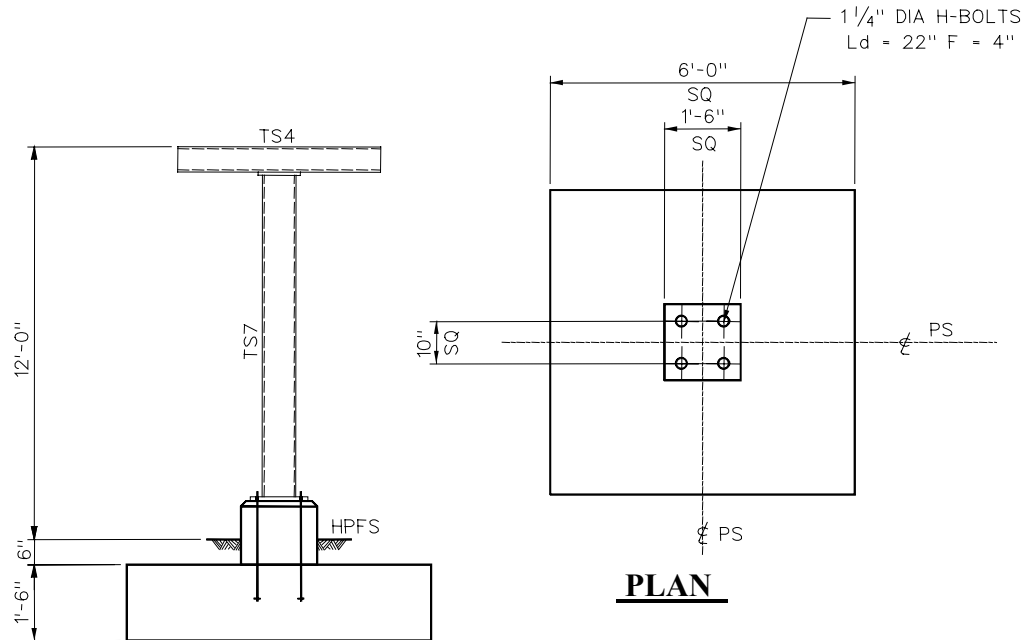
Reinforce for shear, A_v required = 0.2 in^2 per bolt

1 - #4 tie at top of pier is adequate

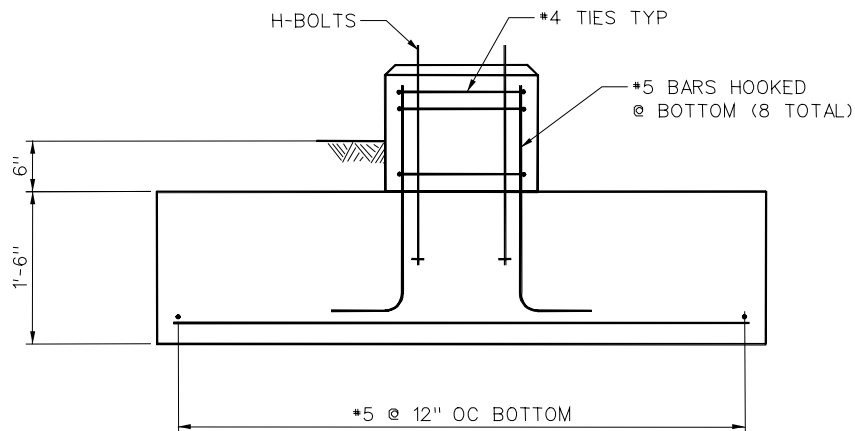
FLUOR

ANCHOR BOLT DESIGN CRITERIA

Sample Design 2 : Steel Column on a Concrete Pier



ELEVATION



TYPICAL SECTION

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Sample Design 2 : Steel Column on a Concrete Pier

Design Data

Concrete $f'_c = 4$ ksi Rebar $f_y = 60$ ksi

Anchors 1-1/4" diameter H-bolts A307 (non-galvanized)

Design basis UBC sect. 1925.3

Reference (Attachment 2)

Working Loads

$P_s = 13.6$ kip per bolt $V_s = 0.48$ kip per bolt

Loads are primarily due to seismic action, therefore friction is excluded.

Factored Loads $c' = 1.3$

$P_u = 13.6(1.3)(1.4) = 24.75$ kip $V_u = 0.48(1.3)(1.4) = 0.87$ kip

Design Anchors

Critical bolt experiences tension and shear

$\phi P_n = 0.9(60 \text{ ksi})(0.77 \text{ in}^2) = 41.58$ kip

refer to (Attachment 3)

$\phi V_n = 0.75(60 \text{ ksi})(0.77 \text{ in}^2) = 34.65$ kip

Interaction:

$$\left(\frac{P_u}{\phi P_n} \right)^2 + \left(\frac{V_u}{\phi V_n} \right)^2 \leq 1.0$$

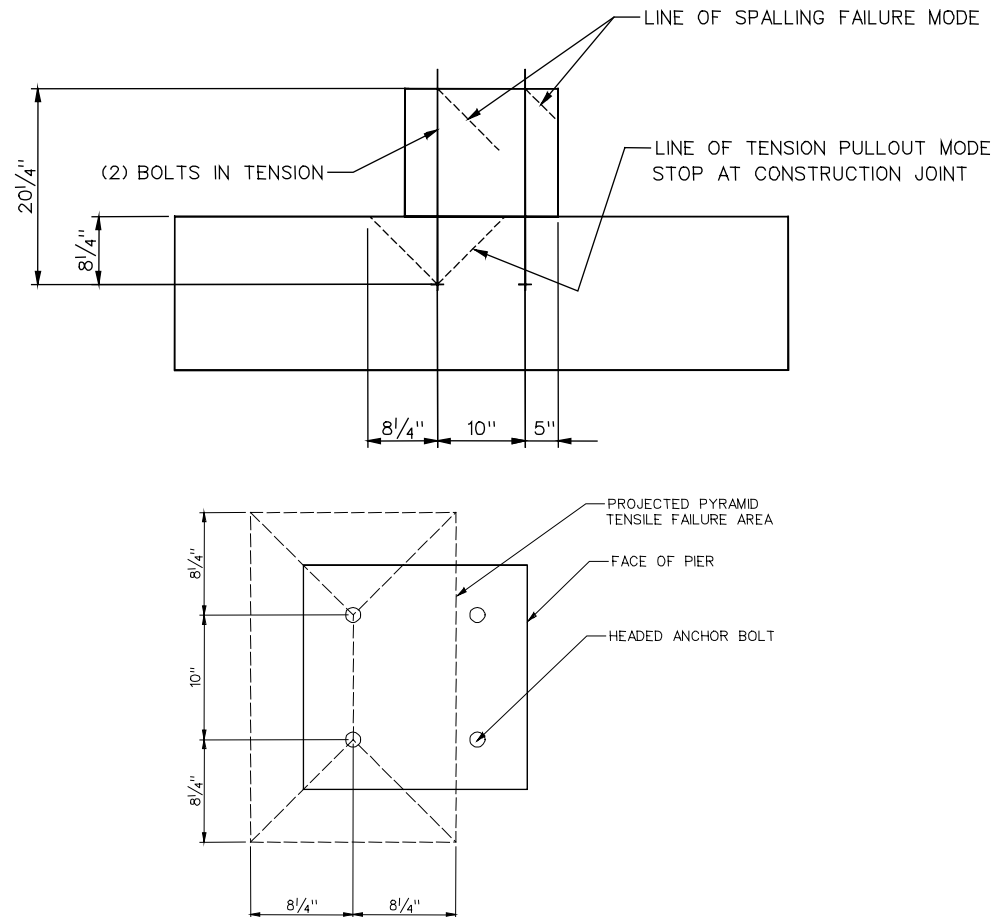
$$\left(\frac{24.75}{41.58} \right)^2 + \left(\frac{0.87}{34.65} \right)^2 = 0.36 \leq 1.0 \quad \text{ok}$$

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Sample Design 2 : Steel Column on a Concrete Pier

Design of Embedment



$$\text{Tension: } A_p = (2) \left(8 \frac{1}{4} \right) \left(8 \frac{1}{4} + 5 \right) = 218.63 \text{ in}^2 \text{ per bolt}$$

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Sample Design 2 : Steel Column on a Concrete Pier

Shear: $A_p = \pi \frac{5^2}{2} = 39.27 \text{ in}^2$

$$\phi P_c = 0.65 (1) 4 \frac{\sqrt{4000}}{1000} (218.63) = 35.95 \text{ kip per bolt}$$

$$\phi V_c = 0.65 (1) 4 \frac{\sqrt{4000}}{1000} (39.27) = 6.46 \text{ kip per bolt}$$

$$P_c = 55.31$$

$$V_c = 9.94$$

Interaction: $\frac{1}{0.65} \left(\left(\frac{24.75}{55.31} \right)^2 + \left(\frac{0.87}{9.94} \right)^2 \right) = 0.35 \leq 1 \text{ ok}$

Equation (3)

$$\text{edge distance} = 5" \leq 10D = 12 \frac{1}{2}"$$

$$\geq \frac{10}{3} D = 4.16"$$

Reinforce for shear

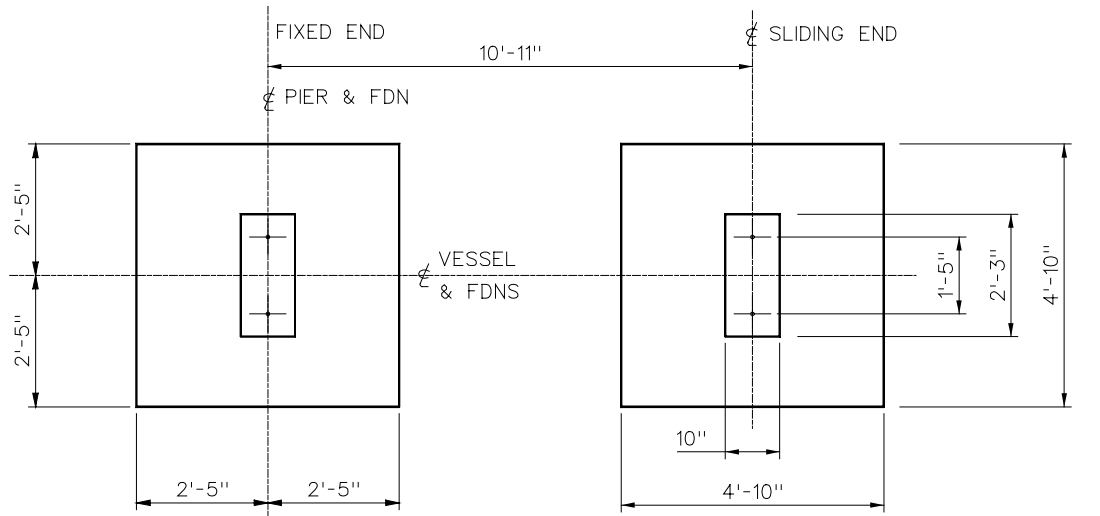
$$A_{sv} = \frac{0.87}{0.9(60\text{ksi})} = 0.02 \text{ in}^2 \text{ per bolt}$$

Typical configuration of #4 ties @ top of the pier is adequate by inspection

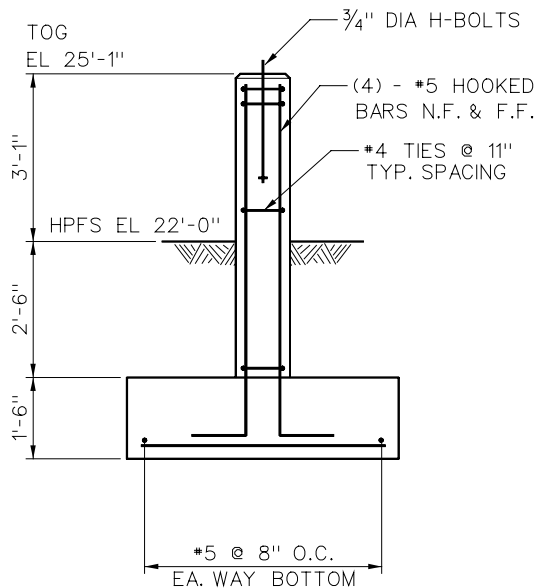
FLUOR

ANCHOR BOLT DESIGN CRITERIA

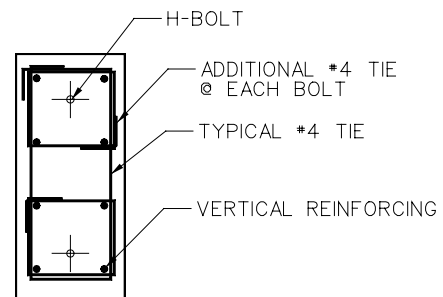
Sample Design 3: Horizontal exchanger on a Concrete Pier



PLAN



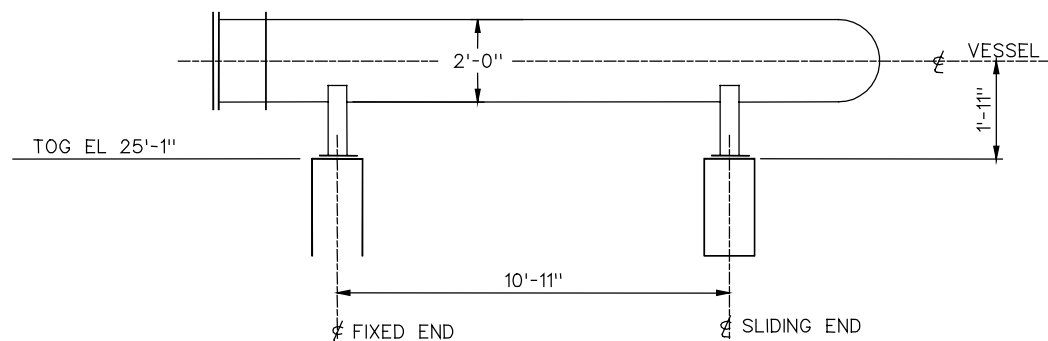
TYPICAL SECTION



TYPICAL PIER DETAIL

ANCHOR BOLT DESIGN CRITERIA

Sample Design 3: Horizontal exchanger on a Concrete Pier



Design Data

Concrete: $f'_c = 4$ ksi $f_y = 60$ ksi

Anchors: A36, 3/4" diameter threaded rods w/nuts, non-galvanized

$$\text{Seismic: } V = \frac{ZIC}{R_w} W_t ; V_{\text{longitudinal}} = 0.275W$$

$$V_{\text{transverse}} = 0.15W$$

Wind and other forces must be checked for typical design

Equipment: Empty $W_t = 4.55$ kip

Operating $W_t = 12.5$ kip

Design per ACI-349 method, seismic design friction excluded.

Loads

Longitudinal seismic operating :

$$V = 0.275(12.5 \text{ kip}) = 3.44 \text{ kip taken @ fixed end}$$

$$M = (3.44 \text{ kip})(1.92') = 6.60 \text{ kip} \cdot \text{ft}$$

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ANCHOR BOLT DESIGN CRITERIA

Sample Design 3: Horizontal exchanger on a Concrete Pier

$$\text{Couple} = 6.6 \text{ ft} - \text{kip}/10.92' = \pm 0.6 \text{ kip } \uparrow \downarrow / \text{end}$$

$$P = 12.5 \text{ kip}/2 = 6.25 \text{ kip/end } \downarrow$$

Loads

Transverse seismic operating:

$$V = 0.275(12.5 \text{ kip}/2) = 1.72 \text{ kip taken @ either end}$$

$$M = (1.72 \text{ kip})1.92' = 3.3 \text{ kip-ft}$$

$$P = 12.5/2 = 6.25 \text{ kip}$$

Tension force on critical bolt = 0.2 kip $\uparrow$, determined by methods beyond the scope of this practice.

Design of Anchors

@ longitudinal forces

$$P_u = 0.75(1.7 \times 1.1)(0.6) = 0.84 \text{ kip per 2 bolts} = 0.42 \text{ kip per bolt}$$

$$V_u = 0.75(1.7 \times 1.1)(3.44) = 4.82 \text{ kip per 2 bolts} = 2.41 \text{ kip per bolt}$$

Steel: 3/4" diameter

$$\phi P_n = 7.13 \text{ kip (from Attachment 1, sheet 1)}$$

$$\phi V_n = 3.70 \text{ kip}$$

Interaction:

$$P_u/\phi P_n + V_u/\phi V_n \leq 1.0$$

$$(0.42/7.13) + (2.41/3.70) = 0.71 \leq 1.0 \text{ ok}$$

Concrete: 3/4" diameter H-bolt $L_d = 12"$

The effective stress area is limited by three edges @ 5" and overlapping stress cones @ 8.5"

ANCHOR BOLT DESIGN CRITERIA

Sample Design 3: Horizontal exchanger on a Concrete Pier

$$A_p \approx (10'')(13.5'') \text{ pyramid failure} = 135 \text{ in}^2$$

$$\phi P_c = 0.65(4) \frac{\sqrt{4000}}{1000} (135) = 22.2 \text{ kip per bolt}$$

$$P' = (60 \text{ ksi})(0.33 \text{ in}^2)^* = 19.8 \text{ kip per bolt} \quad * \text{ use uncorroded or galvanized bolt diameter}$$

$$\phi P_c > P' \quad \text{ok}$$

$$A_p - \text{shear} \approx \pi \left(\frac{5^2}{2} \right) = 39.27 \text{ in}^2$$

$$\phi V_c = 0.65(4) \frac{\sqrt{4000}}{1000} (32.27) = 5.31 \text{ kip per bolt}$$

$$V' = 0.55(60)(0.33 \text{ in}^2) = 10.9 \text{ kip per bolt} < \phi V_c$$

$$\text{edge} = 5'' \leq 10 \left(\frac{3}{4} \right) = 7.5 \quad \text{reinforce}$$

$$A_{sv} = \frac{(0.55)(60)(0.33 \text{ in}^2)}{(0.9)(60 \text{ ksi})} = 0.2 \text{ in}^2 \text{ per bolt}$$

1-#4 tie additional at top of pier around bolts, see sketch

@ transverse loads

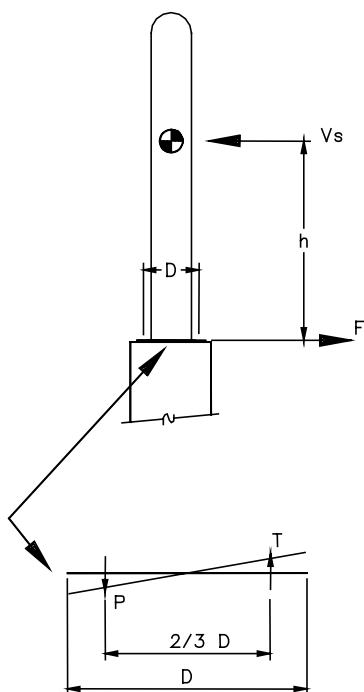
Transverse direction is ok by inspection, since computations show bolt interaction stresses are low and concrete stress values are ok or have already been reinforced.

FLUOR

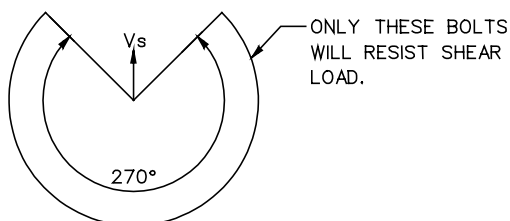
ANCHOR BOLT DESIGN CRITERIA

Friction Resistance (Seismic Loads)

Friction Resistance



$$P = T = \frac{1/4 \times h}{2/3 D}$$



- Assume only those bolts within an arc of 270 degrees, as shown below, resist shear. This way bolts with small edge distances can be ignored.
- When friction resistance at the bottom of the vessel is not sufficient to carry the full lateral force, it is then assumed that the bolts must carry the entire load and friction resistance is zero.

If: $F = P (0.55) \geq V_s$: Bolts do not carry shear load

If: $F = P (0.55) \leq V_s$: Bolts carry full shear load

μ = coefficient of friction = 0.55

Note!!! For non-seismic cases the vertical loads may be used when determining friction resistance.

FLUOR

ANCHOR BOLT DESIGN CRITERIA

Friction Resistance (Seismic Loads)

Design Data

$$V_s = 200 \text{ kip} \quad W = 500 \text{ kip} \quad \mu = 0.55$$

$$M_s = 1500 \text{ ft kip} \quad \text{Concrete: } f'_c = 4 \text{ ksi} \quad f_y = 60 \text{ ksi}$$

$$D = 8'-0" \quad \text{Bolt circle} = 7'-6" \quad \text{ACI design basis}$$

Check Friction Resistance

Unfactored Loads

$$P = \frac{M}{2/3D} = \frac{1500}{2/3(8)}$$

$$= 281.25 \text{ kip}$$

$$F = P \mu = 281.25(0.55) = 154.7 \text{ kip}$$

$$F \leq V_s$$

Factored Loads

$$P_u = \frac{M_u}{2/3D} = \frac{1.4(1500)}{2/3(8)}$$

$$= 393.75 \text{ kip}$$

$$F_u = P_u \mu = 393.75(0.55) = 216.56 \text{ kip}$$

$$F_u \leq V_u$$

Therefore bolts take full shear in both cases (for non-seismic cases, since load factors are different, factored loads may govern the design).

Factored Loads

$$V_u = 0.75(1.1 \times 1.7)200 \text{ kip} = 280 \text{ kip}$$

$$M_u = 0.75(1.1 \times 1.7)1500 \text{ ft kip} = 2100 \text{ ft kip}$$

$$W_u = 0.75(1.4)500 \text{ kip} = 525 \text{ kip}$$

Adjust shear force, V_u to account for 270 degree arc of bolts, thus:

$$V'_u = V_u \left(\frac{360}{270} \right) = 373 \text{ kip}$$

Trial number of bolts = 20

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ANCHOR BOLT DESIGN CRITERIA

Friction Resistance (Seismic Loads)

$$V_u \text{ per bolt} = \frac{373}{20} = 18.7 \text{ kip}$$

Design Tension Force

$$\begin{aligned} P_u &= \frac{4M_u}{N(BC)} - \frac{W}{N} & N &= \text{number of bolts} \\ &= \frac{4(2100)}{20(8)} - \frac{525}{20} \\ &= 26.25 \text{ kips } \uparrow \end{aligned}$$

Design Of Steel Bolts

Refer to Attachment 1, try 2" diameter H-bolts

$$\phi P_n = 69.98 \text{ kip}$$

$$\phi V_n = 36.35 \text{ kip}$$

Interaction:

$$\frac{26.25}{69.98} + \frac{18.7}{36.35} = 0.89 \leq 1.0 \quad \text{ok}$$

Design Of Embedment

A detailed investigation of geometry of stress cones must be done by the design engineer. Overlapping stress cones and intersections of edges are of concern with designs involving equipment that has large, closely spaced bolts.

Also see Attachment 6 for additional design consideration with the design of tall tubular structures.